

Reliability-Screened Routing Under Predictable and Stochastic Travel Variation: Matched Comparisons of Conservative Planning and Statistical Screening

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Abstract

Delivery routes planned on nominal travel times can satisfy every time window on paper yet fail under predictable congestion and random delay. This paper audits a two-layer procedure for the capacitated vehicle-routing problem with time windows. Replaceable OR-Tools generators propose plans, internal window contraction supplies candidate buffers without changing customer promises, and a plan is admitted only when a one-sided 95% Wilson lower bound on the probability that at least 95% of customers are served on time reaches 0.95. Selected plans are then assessed once on untouched scenarios. On 53 seeded synthetic instances with 20–200 customers, three findings emerge. First, on the 46 instances where both exist, the screened procedure exceeds uniformly conservative travel-time padding by 8.3–21.4 percentage points, and for independent disturbances the gap rises from -3.1 to 30.5 points as the log-error scale grows from 0.10 to 0.30. Second, among the 21 instances without an admitted plan, 17 encounter a certified reachability obstruction at one or more tested contraction levels. Third, capping each customer’s contraction at its direct-reachability limit raises admission from 32 to 42 instances and pooled service probability from 78.9% to 86.6%, averaging 97.5–97.6% at 100 and 200 customers, at 2.5% more distance and 0.43 more vehicles per instance. Pooled performance remains below the 95% target. These findings are exploratory, because the same instances informed the design; a prespecified confirmatory protocol for fresh instances accompanies the archive.

Keywords: vehicle routing; time windows; stochastic travel times; chance constraints; simulation-based validation; reproducibility

MSC Classification: 90B06, 90C59, 90C15, 62G09

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1 Introduction

The vehicle-routing problem with time windows (VRPTW) asks a fleet to serve geographically distributed customers subject to capacity and scheduling constraints [13, 15]. In practice, a route that is feasible under one planning-time matrix may miss promised windows when congestion varies or small delays propagate downstream. Minimising nominal distance alone does not price that fragility.

This paper asks four questions:

1. Does the complete screened procedure improve on uniformly conservative planning when both are compared on identical instances?
2. Does any remaining reliability gain arise from route structure or simply from using more vehicles?
3. How do generator choice, clock-profile information and individual contraction caps change admission success and out-of-sample service?
4. How much distance, driver duration, and fleet use are exchanged for the resulting reliability?

The proposed architecture separates *candidate generation* from *stochastic validation*. The optimizer is a nominal VRPTW solver and is not presented as an exact solver for a time-dependent stochastic program. Candidate routes are stress-tested after solving, first on a screening bank and then once on an untouched validation bank. The principal reliability event permits a small daily proportion of late deliveries, avoiding the mechanical collapse of an all-customers-on-time probability as instance size grows.

The contribution is integrative. Time-dependent speed profiles, multi-criteria routing, chance constraints, Monte Carlo evaluation, and Wilson intervals are established ideas. The contribution is a reproducible pipeline that keeps their roles statistically and computationally separate, together with ablations of uniform travel-time padding, clock-profile information, fleet resources, contracted candidates and screening. The screened family and the statistical selection rule act together; a contrast against one uncontracted plan does not identify the selection rule alone.

2 Related work

Solomon’s benchmark study established the standard VRPTW experimental setting [13]; Toth and Vigo provide a broad treatment of vehicle-routing models and algorithms [15]. Time-dependent routing requires care because a naive multiplier applied at departure can violate first-in-first-out (FIFO). Ichoua, Gendreau, and Potvin introduced the integrated speed-profile construction used here [8]; Ghiani and Guerriero later analysed its representational properties [7].

Stochastic VRPTW research has treated uncertain travel and service times using chance constraints, recourse, approximate route-feasibility tests, and data-driven formulations [3, 6, 9, 12]. Ehmke, Campbell, and Urban embed per-customer service levels in a routing method [5]; Taş et al. jointly model time dependence, stochastic travel, service costs, and soft windows [14]; and Bomboi, Buchheim, and Prunte develop route-feasibility checks for correlated and time-dependent travel times [3]. Vareias, Repoussis, and Tarantilis optimize self-imposed time windows under stochastic travel times [16], which is the closest conceptual analogue to the internal window contraction used here.

The present method differs in where the statistical claim is made. It does not embed a stochastic feasibility approximation inside a specialist route-search algorithm. Instead, it accepts routes from a replaceable nominal solver, searches an internal-buffer family on one scenario

bank, and audits the chosen plan once on a held-out bank. Its daily service event constrains the proportion of customers served on time across the whole day; per-customer chance constraints are not used. This solver-agnostic separation is narrower than a new stochastic optimizer, but makes selection failure and out-of-sample reliability directly observable. Monte Carlo approximations to chance constraints require separate assessment of candidate feasibility [1]; the Wilson bound is used because normal intervals behave poorly near probabilities close to one [4, 17].

The implementation uses Google OR-Tools as a nominal candidate generator [10]. It is a practical local-search framework and gives no optimality guarantee. PyVRP provides a substantially stronger hybrid genetic search (HGS) benchmark for canonical VRPTW variants [18]; comparison with such solvers is necessary before interpreting absolute route quality as state of the art.

3 Problem definition

Let $N = \{0, 1, \dots, n\}$ be the nodes, with depot 0 and customers $C = N \setminus \{0\}$. Let V be a homogeneous fleet. Customer i has demand δ_i , deterministic service time s_i , and offered service-start window $[l_i, u_i]$. Vehicle capacity is Q , and the maximum shift duration is $H^{\max}$. The nominal distance and planning time on arc (i, j) are $\bar{d}_{ij}$ and $\bar{t}_{ij}$.

The planning variables are $x_{ijv} \in \{0, 1\}$ for arc use, $z_{iv} \in \{0, 1\}$ for customer assignment, $y_v \in \{0, 1\}$ for vehicle use, service-start time a_{iv} , depot departure d_v , depot return r_v , and slack deficit $G_{iv} \geq 0$. Assignment variables explicitly activate the customer-vehicle copies:

$$\sum_{v \in V} z_{iv} = 1 \quad i \in C, \quad (1)$$

$$\sum_{j \in N \setminus \{i\}} x_{ijv} = \sum_{j \in N \setminus \{i\}} x_{jiv} = z_{iv} \quad i \in C, v \in V, \quad (2)$$

$$\sum_{j \in C} x_{0jv} = \sum_{i \in C} x_{i0v} = y_v \quad v \in V. \quad (3)$$

For delivery-only demand, route capacity can be stated without cumulative-load variables:

$$\sum_{i \in C} \delta_i z_{iv} \leq Q y_v \quad v \in V. \quad (4)$$

The time-propagation constraints exclude customer-only subtours: summing them around any directed customer cycle would imply $0 \geq \sum_i (s_i + \bar{t}_{ij}) > 0$. Positive demands are not needed for that argument. Time windows apply only to assigned copies:

$$l_i z_{iv} \leq a_{iv} \leq u_i z_{iv} \quad i \in C, v \in V. \quad (5)$$

Using separate departure and return variables avoids representing two depot visits with one time variable. With a valid global deactivation constant $M \geq H^{\max} + \max_i s_i + \max_{ij} \bar{t}_{ij}$,

$$a_{jv} \geq d_v + \bar{t}_{0j} - M(1 - x_{0jv}), \quad (6)$$

$$a_{jv} \geq a_{iv} + s_i + \bar{t}_{ij} - M(1 - x_{ijv}), \quad (7)$$

$$r_v \geq a_{iv} + s_i + \bar{t}_{i0} - M(1 - x_{i0v}). \quad (8)$$

The bounds $0 \leq d_v, r_v \leq H^{\max} y_v$ deactivate unused vehicles, and $r_v - d_v \leq H^{\max} y_v$. The implementation enforces these timing relations through OR-Tools Routing dimensions and never builds (8) explicitly, so no big- M relaxation or big- M diagnostic is claimed for the code.

For target slack s^* ,

$$G_{iv} \geq a_{iv} - (u_i - s^*) z_{iv}, \quad (9)$$

$$G_{iv} \geq 0. \quad (10)$$

This makes $G_{iv} = \max\{0, s^* - (u_i - a_{iv})\}$ when $z_{iv} = 1$ and zero when the customer is not assigned to vehicle v .

4 Candidate objectives and scaling

The four raw criteria are

$$F_1 = c_f \sum_{v \in V} \sum_{i,j \in N} d_{ij} x_{ijv}, \quad F_2 = \sum_{v \in V} \sum_{i \in C} G_{iv}, \quad (11)$$

$$F_3 = \sum_{v \in V} (r_v - d_v), \quad F_4 = \sum_{v \in V} y_v. \quad (12)$$

Thus F_3 includes driving, service, and waiting while permitting each vehicle to depart when useful. This removes the fixed-departure artifact that charged a vehicle for depot waiting before its first feasible service.

Single-criterion heuristic solves provide empirical reference values. For each criterion k , let L_k and U_k be the smallest and largest values observed across these anchor solutions. These values are *scaling ranges*, not utopia and nadir points, and they need not contain the Pareto frontier. No value is clipped. The normalized score is

$$Z(\mathbf{x}) = \sum_{k=1}^4 w_k \frac{F_k(\mathbf{x}) - L_k}{\max\{U_k - L_k, 1\}}. \quad (13)$$

The unit fallback prevents division by zero when the anchor archive shows no variation in a discrete criterion. The baseline weights $\mathbf{w} = (0.5693, 0.2643, 0.1055, 0.0609)$ are the principal-eigenvector weights of a single Analytic Hierarchy Process (AHP) pairwise-comparison matrix with consistency ratio 0.0254 [11]; the matrix and calculation appear in Appendix A and in `code/ahp_weights.py`. The matrix was specified for an earlier criteria set in which the third criterion was total distance, and its third row and column were not re-elicited when that criterion became driver duration. The weights therefore express one illustrative preference ordering, fixed before any experiment reported here; they were not elicited from a fleet operator, and sensitivity to them is not evaluated.

OR-Tools implements the distance term as an arc cost, driver duration as a per-vehicle Time-dimension span cost, slack deficit as customer soft upper bounds, and fleet use as a fixed vehicle cost. Integer coefficients are scaled once at the criterion level. The primary planning matrix is nominal and fixed; the original candidate generator does not use clock-dependent costs. The *conservative-speed* baseline replaces 40 km/h by 33.3 km/h throughout the planning matrix. It pads all trips by 20%, including off-peak trips; it does not isolate knowledge of the peak’s timing.

4.1 Clock-aware approximation and deterministic guarantee

A separate distance heuristic freezes each origin’s estimated departure from the preceding schedule, integrates (14) at that time to construct an outgoing travel-time row, and re-solves. Two passes receive 2.5 seconds each. The first reference is the retained nominal plan, whose separate solve cost has already been incurred. Customer departures are estimated service start plus service time; for depot arcs, the estimate is the preceding departure for the first customer of each route, or the mean of used depot departures for other customers. Each resulting sequence is rescheduled with the exact deterministic FIFO profile. Binary search finds its latest feasible whole-minute depot departure; routes failing at departure zero are rejected. The shortest deterministically feasible output is retained, with duration breaking ties. This is an

approximation with explicit failure reporting, not a departure-dependent callback or an exact optimizer for the time-dependent VRPTW.

For a feasible conservative-speed plan, the evaluation speed at $\sigma = 0$ never falls below the planning speed during the planned shift. Starting from the same depot departure, FIFO and induction along each route imply that evaluated service starts and return occur no later than their planned times. Thus deterministic 100% on-time service is guaranteed by construction. The clock-aware approximation's deterministic feasibility is likewise checked before acceptance. These properties are not experimental reliability gains.

5 Time-dependent stochastic evaluation

5.1 FIFO travel times

Partition the day into periods $[T_k, T_{k+1})$ with strictly positive speeds ν_k . The deterministic time-dependent travel time $t_{ij}(\tau)$ is the smallest $\Delta \geq 0$ satisfying

$$\int_{\tau}^{\tau+\Delta} \nu(u) du = d_{ij}, \quad \nu(u) = \nu_k \text{ for } u \in [T_k, T_{k+1}). \quad (14)$$

Because distance accumulated by time is strictly increasing, the arrival map $\tau \mapsto \tau + t_{ij}(\tau)$ is nondecreasing and FIFO holds.

5.2 Correlated disturbances

For scenario ω , arc travel is

$$T_{ij}(\tau, \omega) = t_{ij}(\tau) \Omega_{ij}(\omega), \quad (15)$$

where

$$\Omega_{ij}(\omega) = \exp\{\varepsilon_{ij}(\omega)\}, \quad (16)$$

$$\varepsilon_{ij}(\omega) = \sigma \left(\sqrt{\rho} \xi_0(\omega) + \sqrt{1-\rho} \xi_{ij}(\omega) \right). \quad (17)$$

Here ρ is the correlation between latent Gaussian log-errors. It is not the Pearson correlation between multipliers, which equals

$$\text{Corr}(\Omega_i, \Omega_j) = \frac{\exp(\sigma^2 \rho) - 1}{\exp(\sigma^2) - 1}. \quad (18)$$

The multiplier has median 1 and mean $\exp(\sigma^2/2)$, so the deterministic profile represents the median travel condition, which lies slightly below the mean.

Scenario banks are generated independently of route iteration order. A common factor is drawn once per scenario, while each directed arc receives a deterministic pseudorandom stream keyed by the bank seed and arc endpoints. Plans evaluated on the same bank therefore experience identical disturbances on shared arcs.

6 Screening and independent validation

For a fixed plan R and scenario ω , let

$$q(R, \omega) = \frac{1}{|C|} \sum_{i \in C} \mathbf{1}\{a_i(R, \omega) \leq u_i\} \quad (19)$$

be the on-time customer proportion. The primary service event is

$$J_{\kappa}(R, \omega) = \mathbf{1}\{q(R, \omega) \geq \kappa\}, \quad \kappa = 0.95, \quad (20)$$

and the target is

$$\Pr[J_\kappa(R, \omega) = 1] \geq \alpha, \quad \alpha = 0.95. \quad (21)$$

The stricter zero-lateness fleet-day probability is still reported, but is not the sole admission condition because it declines mechanically as the number of customers increases.

For N scenarios and empirical success $\hat{p}$, the one-sided Wilson lower bound at confidence $1 - \eta$ is

$$L(\hat{p}) = \frac{\hat{p} + z^2/(2N) - z\sqrt{\hat{p}(1 - \hat{p})/N + z^2/(4N^2)}}{1 + z^2/N}, \quad z = z_{1-\eta}. \quad (22)$$

Candidate generation contracts only the internal planning deadline: $[l_i, \max\{l_i, u_i - \beta\}]$. The offered window $[l_i, u_i]$ is unchanged, and every scenario is scored against the original u_i . Candidate values of β are generated with the slack-aware objective and assessed on a screening bank. The anchor ranges are computed once from the uncontracted ($\beta = 0$) instance and held fixed over the buffer search. Among candidates whose screening lower bound reaches α , the one with the smallest normalized planned score (13) is selected. In simulation, every vehicle departs at its planned d_v ; there is no scenario-dependent departure recourse. The selected plan is evaluated once on a larger, untouched validation bank. This separation prevents adaptive candidate search from selecting on the evidence used for final reporting.

For procedure-level reporting, a prespecified fallback makes the output defined on every instance. If no candidate passes screening, the uncontracted slack-aware plan ($\beta = 0$) is returned. Validation never changes this choice; it only estimates the performance of the plan already chosen. Selected-only results are shown only when labelled as such; the primary procedure comparison includes all instances.

6.1 Generator replacement and reachability-capped contraction

Screened conservative-speed uses the same buffer grid, admission bank and Wilson threshold, but generates distance-minimizing plans with the padded matrix. Among passing plans it chooses the shortest distance, breaking ties by planned duration and then buffer. Its fallback is the uncontracted conservative plan; if that plan was not found, the retained uncontracted slack-aware plan is used. This makes the replacement procedure defined on all 53 instances while preserving the base generator’s 46/53 found-plan count.

Let $e_i = \max\{l_i, \bar{t}_{0i}\}$ be earliest direct service under the nominal matrix. The capped weighted variant uses the internal upper bound

$$u_i^{\text{cap}}(\beta) = \min\{u_i, \max\{l_i, u_i - \beta, e_i\}\}. \quad (23)$$

This prevents an initially reachable customer from becoming unreachable by a direct trip solely because of contraction. It does not imply joint routing feasibility or the admission target. Identical contracted models reuse their archived candidates; only changed models are re-solved. Selection still uses the fixed original scaling ranges and weighted score, with the same fallback.

For diagnosis, a shortest-path relaxation uses arc cost $\bar{t}_{ij} + s_i$ (with $s_0 = 0$), allowing intermediate customers without capacity or window restrictions. Its depot-to- i distance is a valid lower bound even if rounded distances violate a triangle inequality. A contracted deadline below this bound certifies infeasibility. Without a certificate, a missing solver output is labelled *no feasible plan found*; neither a time limit nor passing the reachability test proves feasibility. Solver status certificates were not retained in the archived runs and are not inferred retrospectively.

7 Experimental design

Synthetic instances place a central depot and uniformly distributed customers on a 100×100 grid. Demand is uniform on 10–49 units, service time is 10 minutes, window width is uniform

on 90–149 minutes, capacity is 1000, and the shift limit is 600 minutes. Time zero is 06:00 local time, so simulated clock time is obtained by adding 360 minutes to each relative schedule time. The retained design uses the seeded sizes in Table 1; all model and simulation constants are disclosed in Table 2.

Table 1 Seeded synthetic experimental design.

Customers	Vehicles available	Instances	Seeds
20	6	20	7200–7219
50	14	15	7500–7514
100	26	10	8000–8009
200	40	8	9000–9007

Table 2 Planning, disturbance, and statistical parameters.

Parameter	Retained value
Nominal/free-flow speed $\bar{v}$	40 km/h
Shift and clock origin	600 min, beginning 06:00 local time
Capacity; service time; target slack s^*	1000; 10 min; 15 min
Distance cost c_f	15 cost units/km
AHP weights (w_1, w_2, w_3, w_4)	(0.5693, 0.2643, 0.1055, 0.0609)
Integer objective scale	100,000
Planned-shift speed divisors	0.9 (06:00–07:00), 1.2 (07:00–09:00), and 1.0 (09:00–16:00); speed is $\bar{v}$ divided by the divisor
Outside-shift continuation	1.0 to 17:00, 1.3 (17:00–19:00), 0.9 thereafter; used if a stochastic route overruns the shift
Log-error scale σ ; latent correlation ρ	0.20; 0.50
Exploratory disturbance grid	$\sigma \in \{0.10, 0.20, 0.30\}$; $\rho \in \{0, 0.5, 0.9\}$
Service fraction κ ; target α	0.95; 0.95
Wilson tail probability η	0.05 (one-sided 95% lower bound)
Screening; validation; sensitivity scenarios	1000; 5000; 1000 per cell
Buffer grid	$\beta \in \{0, 5, \dots, 60\}$ min

Eight configurations are compared:

1. **Nominal distance:** fixed 40 km/h planning matrix and a distance-minimizing objective.
2. **Conservative-speed distance:** distance minimization with a uniform 33.3 km/h planning matrix.
3. **Slack-aware:** objective (13) with $\beta = 0$, isolating the objective from screening.
4. **Fleet-matched distance:** distance minimization forced to use exactly the fleet count of the slack-aware plan and initialized from its feasible routes.
5. **Original procedure:** the complete weighted internal-buffer search, with selection on 1000 screening scenarios, the prespecified $\beta = 0$ fallback, and assessment on 5000 fresh scenarios.
6. **Screened conservative:** conservative-speed distance candidates with the same admission test and the fallback defined above.
7. **Capped procedure:** the original weighted procedure with customer-specific reachability caps.

8. **Clock approximation:** two departure-estimate matrix updates and exact deterministic route rescheduling.

The nominal shift ends at 16:00. The full-day speed profile remains defined beyond that time because lognormal delays are unbounded and a stochastic route can overrun. Evening periods lie outside the planned shift but remain in the evaluator. Fleet-matched distance is initialized from the slack-aware routes; because this can in principle transfer their structure, the initialization is treated as part of the comparator’s definition.

The screening grid is $\beta \in \{0, 5, \dots, 60\}$ minutes. The four single-criterion anchor solves use two seconds each to stabilize the scaling ranges; reported candidate configurations use a five-second OR-Tools search budget for each nominal solve. If a candidate solve returns no feasible plan, the same five-second call is attempted at most twice more; the first feasible result is kept, and no best-of-several selection is made. Reachability-certified new models need no solver attempt. The clock approximation instead divides a total five-second budget between its two solves. Six experiment workers ran in parallel; elapsed time and software metadata are retained. All configurations use the same fresh validation bank within an instance. The reported reliability outcomes are service-event probability, zero-lateness fleet-day probability, mean on-time fraction, expected number late, mean and 95th-percentile total lateness, and mean route success. Operational outcomes are distance, driver duration, and fleet use. Simulated duration is additionally reported under the common evaluation model, to distinguish actual route time from each generator’s planned F_3 . Differences are paired by instance. Confidence intervals across instances are distinguished from binomial Monte Carlo intervals within an instance. Across-instance intervals use a paired nonparametric bootstrap with 20,000 resamples and a fixed analysis seed. Claims of distance equivalence require a pre-specified equivalence margin; failure to reject a difference is not treated as proof of equality.

8 Results

8.1 Complete procedures on matched instances

Table 3 puts every original configuration on the same conservative-feasible subset. The paired contrasts are computed from individual instance differences. Their intervals therefore quantify the comparison directly; they are not obtained by subtracting endpoints of separate confidence intervals.

Table 3 Validated service-event probability (%) on the same 46 instances: every entry in a row has the displayed denominator. Entries are mean [95% instance-bootstrap interval]. N, C, S, F and P denote nominal, conservative-speed, slack-aware, fleet-matched and original procedure, respectively.

n	k	N	C	S	F	P
20	18	21.5 [18.8, 24.4]	64.2 [61.1, 67.4]	53.3 [47.6, 59.4]	21.7 [18.9, 24.8]	85.7 [74.4, 95.0]
50	13	17.7 [16.1, 19.6]	59.0 [54.6, 63.1]	45.8 [43.1, 48.4]	16.9 [14.8, 18.9]	76.8 [62.3, 89.7]
100	8	25.4 [23.5, 27.1]	71.6 [69.1, 74.0]	55.3 [51.5, 59.3]	26.2 [23.6, 29.0]	87.6 [74.2, 97.5]
200	7	26.7 [24.9, 28.8]	74.6 [72.2, 77.0]	61.7 [58.3, 66.8]	27.7 [25.5, 30.6]	82.9 [69.3, 94.3]

In the paired contrasts of Table 4, the complete original procedure improves on conservative-speed planning by 8.3–21.4 percentage points across sizes. The unscreened slack-aware objective does not show this advantage. At $n = 200$ the interval includes zero, so the positive mean at that size is not conclusive.

Table 4 Paired probability differences (percentage points) on exactly the same subsets as Table 3. SC denotes screened conservative-speed with its prespecified fallback.

n	S minus C	P minus C	SC minus C
20	-11.0 [-16.1, -6.0]	21.4 [9.6, 31.3]	28.2 [21.4, 34.3]
50	-13.1 [-17.4, -8.9]	17.9 [2.8, 32.0]	11.8 [2.7, 22.2]
100	-16.2 [-20.6, -11.7]	16.0 [4.8, 24.8]	11.2 [2.8, 19.4]
200	-12.9 [-18.5, -5.7]	8.3 [-5.1, 20.0]	5.7 [0.0, 12.9]

Table 5 Paired original-procedure changes relative to conservative speed on the common subset. Distance and duration are mean percentage changes; fleet is an absolute difference. Scheduled duration is each planner’s own F_3 . Simulated duration uses the same realized traffic model and planned departures for both configurations.

n	Distance	Scheduled time	Simulated time	Fleet
20	4.9 [2.4, 7.3]	0.3 [-3.8, 4.8]	8.5 [4.4, 12.9]	0.22 [0.00, 0.44]
50	2.3 [-0.6, 5.2]	-7.5 [-12.2, -2.7]	-2.0 [-7.0, 2.8]	-0.46 [-1.00, 0.00]
100	2.0 [-1.3, 5.1]	-3.0 [-9.4, 2.5]	2.2 [-4.3, 7.6]	0.12 [-0.38, 0.62]
200	2.4 [-1.4, 5.7]	-7.7 [-15.2, -1.0]	-2.8 [-10.6, 4.1]	-0.71 [-2.57, 1.00]

8.2 Resources on the common subset

Scheduled driver duration is not identical to realized labour time: the conservative planner budgets padded arc times, while the evaluator may traverse those arcs faster. Table 5 therefore includes both measures. Absolute distance premiums also carry solver-quality uncertainty: the five-second deterministic OR-Tools benchmark remains about 11% longer than HGS at $n = 200$.

The matched data do not support uniform resource dominance: at $n = 20$ the original procedure has slightly greater scheduled time and more vehicles; at $n = 100$ it also uses slightly more vehicles. The simulated-time interval excludes zero only for the increase at $n = 20$. The service gain is therefore a trade-off against some operating resources.

8.3 Screening a replaceable generator and capping contractions

Table 6 All-instance procedure outcomes. Selected counts admission-bank lower bounds reaching 0.95; retained counts those selections whose fresh validation lower bound also reaches 0.95. Probability includes every fallback, as well as selected plans.

n	Procedure	Selected	Retained	Probability (%)
20	Original procedure	14/20	14/14	81.6 [70.2, 91.9]
20	Screened conservative	15/20	15/15	87.7 [79.0, 94.7]
20	Capped procedure	14/20	14/14	81.6 [70.3, 92.0]
50	Original procedure	8/15	8/8	72.7 [59.2, 86.4]
50	Screened conservative	4/15	4/4	67.5 [58.2, 77.3]
50	Capped procedure	10/15	10/10	80.1 [66.9, 90.8]
100	Original procedure	6/10	5/6	81.3 [68.8, 93.3]
100	Screened conservative	4/10	4/4	77.5 [67.2, 88.2]
100	Capped procedure	10/10	9/10	97.5 [96.7, 98.2]
200	Original procedure	4/8	3/4	80.6 [68.7, 92.3]
200	Screened conservative	2/8	2/2	78.3 [71.0, 86.5]
200	Capped procedure	8/8	7/8	97.6 [96.9, 98.2]

In Table 6, the screened conservative generator selects on 25/53 instances; 25 selections retain the target lower bound. Its all-instance paired difference from the original procedure is

-0.3 [-5.5, 5.0] points. The capped variant selects on 42/53, with 40 retained; its corresponding difference is 7.7 [3.5, 12.5] points. Capping removes only an individual reachability obstruction; capacity, sequencing and finite-search limitations remain.

Table 7 Operational price of the capped variant relative to the original procedure on all 53 instances. Percentage changes are paired per instance; fleet changes are absolute. Capping can permit different route and resource choices, so its reliability gain is not attributed to schedule structure alone.

n	Distance	Scheduled time	Simulated time	Fleet
20	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.00 [0.00, 0.00]
50	2.2 [0.0, 5.4]	4.6 [0.0, 11.4]	4.5 [0.0, 11.3]	0.33 [0.00, 0.87]
100	4.9 [1.1, 8.9]	6.8 [1.7, 12.1]	6.6 [1.7, 11.8]	0.30 [-0.20, 0.80]
200	6.3 [1.6, 10.9]	10.5 [3.1, 18.0]	10.0 [2.9, 17.1]	1.88 [0.50, 3.38]

Pooled over all instances (Table 7), the capped variant costs 2.5% more distance and 0.43 additional vehicles than the original procedure. At $n = 200$, the additional fleet use is material and should not be obscured by the earlier fleet-matched ablation of the uncontracted objective.

8.4 Clock-aware approximation

Table 8 Clock-aware approximation: identical matched subsets within each row, restricted to an exact deterministic schedule being found by the approximation. Probabilities and paired changes are percentages and percentage points, respectively. The two matrix solves share a five-second search allowance.

n	Found	Nominal	Clock-aware	Original P	Clock minus N
20	19/20	21.2 [18.6, 24.0]	39.5 [37.9, 41.3]	84.1 [73.2, 93.6]	18.4 [15.1, 21.4]
50	14/15	17.5 [16.0, 19.3]	36.5 [35.4, 37.7]	74.8 [60.8, 87.1]	19.0 [17.2, 20.9]
100	10/10	25.4 [23.9, 26.8]	44.5 [43.0, 46.0]	81.3 [68.8, 93.2]	19.1 [17.6, 20.8]
200	6/8	26.6 [24.5, 28.9]	43.7 [42.3, 45.3]	81.7 [68.7, 93.6]	17.2 [15.7, 18.7]

The clock-aware rows test a practical departure-time approximation. Exact deterministic retiming verifies each reported route, so its success at $\sigma = 0$ is an acceptance condition and says nothing about stochastic robustness. Conservative-speed planning likewise guarantees deterministic on-time service within the planned horizon by upper-bounding every arc time. Both 100% rows hold by construction.

In Table 8, the approximation improves on nominal distance by 17.2–19.1 points on its feasible subsets, but remains below the original procedure. Its latest-feasible-departure rule and two-pass search are part of this result; the comparison does not establish what an optimal time-dependent planner could achieve.

8.5 Matched disturbance sensitivity

The mechanism comparison of interest is the paired P-minus-C column of Table 9, which Fig. 1 also plots across the disturbance grid. Changes in its sign or magnitude describe fixed-plan performance under each setting. On its own, this column cannot separate screening from the contracted candidate family, and the nine cells share instances and common random numbers.

For independent disturbances, increasing σ from 0.10 to 0.30 changes the paired contrast from -3.1 [-10.7, 3.9] to 30.5 [19.8, 40.5] points. This supports a growing advantage of the complete screened family in that regime. At $\rho = 0.9$ the increase is not monotonic, so the data do not support a universal monotonic relationship between noise scale and screening value.

Table 9 Fixed-plan sensitivity on the common 46 instances. Each cell uses 1,000 common scenarios from a bank separate from primary validation; plans are not re-screened or reoptimized for each parameter setting. C, P and SC are conservative speed, original procedure and screened conservative.

σ	ρ	C (%)	P (%)	SC (%)	P minus C (points)
0.10	0.0	91.8 [88.2, 95.0]	88.8 [81.3, 95.3]	97.1 [94.3, 99.2]	-3.1 [-10.7, 3.9]
0.10	0.5	84.6 [81.5, 87.4]	87.8 [81.5, 93.5]	93.2 [90.1, 95.9]	3.2 [-3.3, 9.3]
0.10	0.9	81.2 [78.3, 84.0]	87.6 [81.4, 93.3]	91.5 [88.2, 94.5]	6.4 [0.0, 12.5]
0.20	0.0	69.8 [64.1, 75.5]	81.0 [71.4, 89.7]	86.7 [81.0, 92.0]	11.2 [1.4, 20.6]
0.20	0.5	65.6 [63.2, 67.9]	82.8 [75.9, 89.2]	82.9 [78.0, 87.5]	17.2 [10.3, 23.7]
0.20	0.9	67.2 [65.3, 68.9]	82.5 [76.8, 87.9]	82.1 [77.9, 86.2]	15.4 [9.5, 20.9]
0.30	0.0	41.7 [37.4, 46.0]	72.1 [61.1, 82.4]	69.4 [60.6, 77.8]	30.5 [19.8, 40.5]
0.30	0.5	54.4 [52.5, 56.3]	74.0 [67.8, 79.8]	72.1 [67.1, 77.0]	19.6 [13.6, 25.3]
0.30	0.9	60.6 [59.3, 61.9]	75.6 [70.7, 80.1]	74.4 [70.6, 78.2]	15.0 [10.0, 19.6]

8.6 Diagnosis of screening failures

Of the 21 original screening failures, 17 reach a buffer with an explicit customer-reachability contradiction and 4 have no such certificate through 60 minutes. 8 instances also contain a no-plan outcome before any certificate. Those outcomes remain unresolved: failure of a necessary reachability test proves infeasibility, but passing it does not establish a feasible multi-customer routing. The per-instance audit appears in Appendix B.1.

An additional exploratory check took the 8 failures whose search ended at a no-plan outcome without a reachability certificate and re-solved exactly that contracted model with a longer budget: 30 s, then 120 s for those still without a plan. In 4 cases a plan was found at that contraction level, so the original outcome was a search limit. The remaining 4 still have no plan after 120 s, which does not prove infeasibility. The script and records are part of Online Resource 1.

Appendix B reports customer outcomes on the matched subset (Table 10), absolute planned resources (Table 11), service-threshold sensitivity (Table 12), the instance-level failure audit (Table 13), and the equal-budget deterministic benchmark against PyVRP/HGS (Table 14).

9 Reproducibility and implementation checks

The study archive (Online Resource 1) contains the candidate generator, scenario-bank implementation, test suite, experiment driver, raw JSON records, and the script that produces the manuscript tables and figure. Tests cover FIFO behavior, lognormal moments, order-invariant common random numbers, Wilson calculations, visit and capacity feasibility, and the requirement that contracted candidates are scored against original windows. Runtime metadata record Python, NumPy, OR-Tools, operating system, solver budget, seeds, and scenario counts.

The clock convention is essential for reproduction: relative time zero is 06:00 and the evaluator adds 360 minutes before entering the full-day speed profile. Omitting that offset instead evaluates a 00:00–10:00 shift. The vectorized evaluator is checked against the original scalar implementation including period boundaries, customer lateness and shared arc disturbances.

The revision protocol was fixed before running the new candidates. Admission uses seed 100000+ instance seed; all fixed outputs are assessed with 5000 fresh scenarios from seed 400000+ instance seed. The 1000-scenario disturbance sweep uses seed 500000+ instance seed. Earlier aggregate results motivated these analyses, so they form an exploratory extension on known synthetic instances. They are neither preregistered nor an external replication. The fresh draws do not remove reuse of the instance set. Plans in the sweep are not re-screened when σ or ρ changes. A proposed confirmatory protocol specifies four directional hypotheses, their tests, decision rules and failure policy for 53 new instances whose seeds are shifted by 60000. It

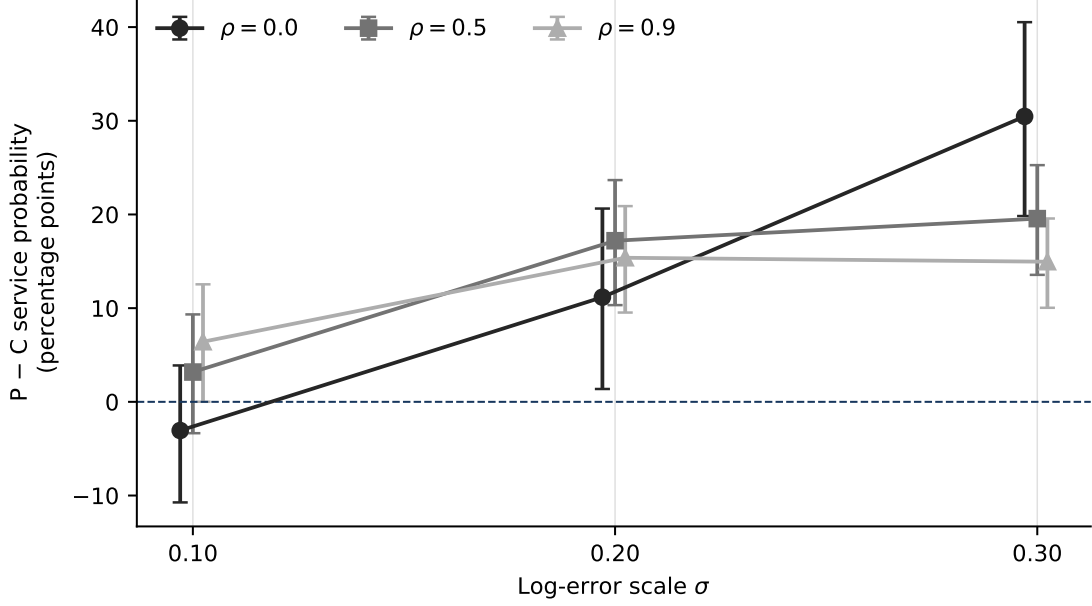


Fig. 1 Paired difference in service probability between the original procedure (P) and conservative-speed planning (C) on the common 46 instances. Circles, squares and triangles mark $\rho = 0$, 0.5 and 0.9, and the dashed line marks zero. Faint vertical lines mark $\sigma = 0.10$, 0.20 and 0.30; markers are offset slightly around them for legibility only. Error bars are 95% bootstrap intervals over instances. Plans are fixed across this disturbance sweep, and the intervals are not simultaneous across cells

becomes fixed when publicly timestamped with a hash manifest of the code that runs and analyses it. One pipeline smoke test, on an instance outside that set, was run while the protocol was prepared and is disclosed in it as pilot data. The protocol’s results are not part of this manuscript.

Every reported table and figure is regenerated from retained machine-readable records. Internal planning buffers are always evaluated against offered windows, candidate selection and final assessment use independent banks, and solver-quality comparisons use equal wall-clock budgets and identical deterministic instances. These rules prevent implementation diagnostics or adaptive within-run selection from using its own validation outcomes.

Use of generative AI tools. Generative AI assistants were used for code development, experiment implementation, code review, analysis checking, and drafting and editing text. The assistants were OpenAI’s ChatGPT (Astra 6) and Anthropic’s Claude (Opus 5, Opus 5.5 and Sonnet 5). They are not authors. Every experiment was run from the archived code, every reported number regenerates from the raw records, and the headline results were recomputed independently of the analysis scripts. The author reviewed all code and text and takes full responsibility for the content.

10 Limitations

First, the generators use nominal or iteratively frozen matrices, so the procedure can miss better stochastic plans. The candidate generator is also below the state of the art: under an equal five-second budget, OR-Tools returns routes about 11% longer than PyVRP/HGS at 200 customers (Table 14). Absolute distance premiums therefore carry solver quality, and some of the uncertified screening failures may reflect search limits. Conservative speed is uniform padding,

and the clock-aware approximation is not an exact time-dependent optimizer. Generator and objective choices also differ between screened families; the experiment is not a full factorial decomposition of every design decision. Second, the disruption parameters are assumed and have not been calibrated to fleet GPS residuals, so the results hold conditionally on those values and are not operational forecasts. An exploratory σ - ρ grid is reported, but it does not replace local calibration. Third, the geometry is synthetic and the deterministic solver check is not a substitute for Solomon or Gehring–Homberger benchmarks or a real road network. Fourth, the AHP weights are one preference vector and require global sensitivity analysis. Fifth, Wilson screening addresses binomial sampling uncertainty for a fixed candidate; its nominal coverage is not a simultaneous guarantee across an adaptively searched grid. Held-out assessment is therefore necessary. It is also not a proof of feasibility under distributional misspecification. Sixth, breakdown recovery, heterogeneous fleets, new-order insertion, and rolling-horizon replanning are excluded from the present evidence.

11 Conclusion

The evidence separates three mechanisms. Uniformly conservative travel times protect against predictable congestion by construction, padding every trip including off-peak ones. On matched instances the complete screened procedure nevertheless exceeds them by 8.3–21.4 percentage points on average, although the interval at 200 customers includes zero; for independent disturbances its advantage grows with noise, from -3.1 to 30.5 points. The growth is not monotonic under strong correlation, so no universal ordering is claimed. The unscreened slack-aware objective alone falls below conservative padding. The advantage comes from the complete procedure, in which the contracted candidate family and statistical selection act together.

Screening failures can be diagnosed. Among the 21 instances without an admitted plan, 17 encounter a certified reachability obstruction at one or more tested contraction levels. A certificate proves that a particular contracted model is infeasible; it does not show that reachability alone caused the procedure to miss its target. Capping each customer’s contraction at its direct-reachability limit raises admission from 32 to 42 instances. Of the instances with 100 or 200 customers, 18/18 were admitted and 16/18 retained the target lower bound under fresh validation, with mean service probability 97.5–97.6%. The gain costs 2.5% more distance and 0.43 more vehicles on average, rising to 1.88 more vehicles at 200 customers. The earlier fleet-matched experiment concerns the uncontracted objective and does not isolate the capped variant’s mechanism. At 20 customers capping changes nothing; only two of the six failures there encounter a certificate. Of the other four, a longer search budget finds a plan for two, and two remain unresolved.

Pooled over all instances, no procedure reaches the 95% target, so procedure averages and selected-plan compliance must both be reported. The shortfall comes from instances without an admitted plan. Plans admitted by screening average 97.3% for the original procedure and 97.4% for the capped procedure, while the fallback plans used elsewhere average 50.9% and 45.2%. All 11 capped-procedure fallbacks occur at 20 or 50 customers, so closing the remaining gap depends on admitting plans for small instances. These results are conditional on assumed disruption parameters and synthetic geometry, and they are exploratory: the instances that produced them also motivated the design. The prespecified confirmatory protocol states in advance what fresh instances must show for the two primary findings to stand.

Statements and Declarations

Competing interests. The author has no relevant financial or non-financial interests to disclose.

Funding. No funding was received for conducting this study.

Data and code availability. The reproducibility archive contains source code, dependency versions, tests, seeds, raw JSON records, analysis scripts, generated tables and figures, and the confirmatory protocol. It is supplied as Online Resource 1 and archived at [\[Zenodo DOI\]](#) [2].

Use of generative AI. Described in Section 9.

Ethics approval. Not applicable: the study uses synthetic data only.

Consent. Not applicable.

Author contributions. S.A. is the sole author and conceived the study, implemented the software, ran the experiments, and wrote the manuscript.

A AHP pairwise-comparison matrix

Criteria are ordered as distance cost, slack deficit, driver duration and fleet use. Entry a_{kl} is the judged importance of criterion k relative to l on Saaty’s 1–9 scale:

$$\mathbf{A} = \begin{pmatrix} 1 & 3 & 5 & 7 \\ 1/3 & 1 & 3 & 5 \\ 1/5 & 1/3 & 1 & 2 \\ 1/7 & 1/5 & 1/2 & 1 \end{pmatrix}.$$

Its principal eigenvalue is $\lambda_{\max} = 4.0685$, and the normalised principal eigenvector is

$$\mathbf{w} = (0.5693, 0.2643, 0.1055, 0.0609).$$

With Saaty’s random index 0.90 for four criteria, the consistency index is $(4.0685 - 4)/3 = 0.0228$ and the consistency ratio is 0.0254. As stated in Section 4, the third row and column were set when the third criterion was total distance and were not re-elicited for driver duration.

B Additional matched outcomes and failure records

Table 10 Customer outcomes on the conservative-feasible subset; the clock approximation is restricted further when needed. Counts make this additional conditioning explicit.

Configuration	k	On time (%)	Expected late
Nominal	46	83.6 [82.5, 84.6]	10.00 [7.95, 12.20]
Conservative speed	46	93.9 [93.4, 94.4]	3.47 [2.84, 4.14]
Slack-aware	46	90.9 [90.0, 91.7]	5.46 [4.38, 6.65]
Fleet-matched	46	83.0 [81.9, 84.0]	10.47 [8.29, 12.87]
Original procedure	46	96.6 [95.1, 97.9]	2.20 [1.26, 3.31]
Screened conservative	46	97.1 [96.3, 97.8]	2.20 [1.47, 2.99]
Capped procedure	46	97.7 [96.4, 98.9]	0.91 [0.56, 1.32]
Clock approximation	44	88.0 [87.2, 88.7]	6.73 [5.39, 8.23]

Table 11 Absolute planned resource means on the same subset as Table 10: distance in km, scheduled duration in minutes, and vehicles. These are not optimality claims.

Configuration	k	Distance	Duration	Fleet
Nominal	46	1180.7 [989.9, 1381.0]	2906.5 [2379.9, 3481.3]	7.67 [6.46, 8.98]
Conservative speed	46	1216.2 [1021.5, 1422.0]	3352.5 [2738.5, 4011.0]	8.30 [6.93, 9.76]
Slack-aware	46	1196.3 [1003.0, 1404.0]	2902.0 [2380.3, 3466.2]	7.46 [6.22, 8.78]
Fleet-matched	46	1174.2 [989.2, 1372.8]	2833.6 [2330.6, 3382.6]	7.46 [6.22, 8.76]
Original procedure	46	1247.9 [1049.3, 1453.0]	3171.2 [2620.9, 3777.3]	8.17 [6.89, 9.57]
Screened conservative	46	1252.8 [1056.4, 1461.2]	3494.5 [2874.0, 4161.0]	8.76 [7.35, 10.28]
Capped procedure	46	1283.5 [1076.5, 1506.8]	3317.3 [2709.4, 3964.1]	8.52 [7.13, 10.04]
Clock approximation	44	1137.2 [955.5, 1330.9]	2868.3 [2335.0, 3456.8]	7.64 [6.32, 9.11]

Table 12 Service-threshold sensitivity on the common 46; columns use the same plans and fresh primary bank. At $n = 20/50/100/200$, allowed-late counts are 2/5/10/20 at $\kappa = .90$, 1/2/5/10 at .95, 0/1/2/4 at .98, and zero at 1.00.

κ	N	C	S	P	SC
0.90	38.8 [36.5, 41.2]	80.6 [78.8, 82.4]	68.8 [65.9, 71.7]	89.2 [84.2, 93.7]	91.2 [88.5, 93.8]
0.95	21.9 [20.3, 23.5]	65.6 [63.1, 68.0]	52.8 [49.9, 55.8]	83.1 [76.3, 89.4]	82.8 [77.9, 87.6]
0.98	11.2 [10.3, 12.2]	48.2 [45.6, 50.8]	36.4 [33.8, 39.0]	75.9 [68.2, 83.3]	73.2 [67.1, 79.3]
1.00	6.4 [5.4, 7.4]	34.4 [31.8, 37.0]	26.1 [23.8, 28.5]	68.2 [59.6, 76.2]	62.4 [53.8, 70.9]

B.1 Instance-level failure audit

Table 13 Every original screening failure. Best buffer and lower confidence bound (LCB) refer to a returned candidate; first no-plan is an observed solver outcome; first proof is the earliest grid point with a reachability certificate. Unresolved indicates at least one no-plan result without that certificate. A dash means no certificate through 60 minutes.

n	Seed	Best β	LCB (%)	No-plan β	Proof β	Unresolved
20	7200	45	90.7	50	–	Yes
20	7201	35	93.5	40	–	Yes
20	7210	35	90.8	45	–	Yes
20	7212	5	37.9	10	–	Yes
20	7216	55	90.7	60	60	No
20	7218	10	65.9	15	50	Yes
50	7501	25	87.5	30	35	Yes
50	7503	10	64.1	15	15	No
50	7504	35	91.3	40	40	No
50	7506	0	44.7	5	5	No
50	7508	35	85.9	40	40	No
50	7511	35	88.5	40	50	Yes
50	7513	40	93.6	45	60	Yes
100	8003	20	85.3	25	25	No
100	8004	15	81.5	20	20	No
100	8007	5	73.6	15	15	No
100	8008	25	91.7	35	35	No
200	9004	10	71.2	15	15	No
200	9005	20	88.9	25	25	No
200	9006	25	94.9	40	40	No
200	9007	20	86.4	25	25	No

Table 14 Retained equal-budget deterministic-core benchmark: OR-Tools excess distance relative to PyVRP/HGS (%). Counts are valid one-second/five-second pairs. There was one missing pair at the one-second budget.

n	Valid pairs (1s/5s)	One second	Five seconds
20	19/20	1.0 [0.3, 1.8]	0.0 [0.0, 0.0]
50	15/15	4.8 [3.3, 6.2]	3.0 [1.9, 4.2]
100	10/10	8.5 [5.9, 11.1]	6.3 [4.7, 7.9]
200	8/8	32.9 [29.4, 36.2]	11.1 [9.3, 13.5]

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